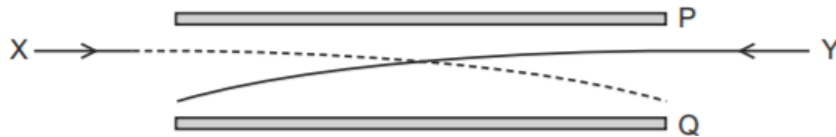


so, $b = \frac{\lambda}{\theta} = 600 \times 10^{-9} \times 6.3 \times 10^{11} = 3.8 \times 10^5 \text{ m}$

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The diagram shows the paths of two charged particles, X and Y, during their passage between a pair of oppositely charged metal plates, P and Q.



The plates are charged such that the electric field between them is directed from P to Q.

Which charges on X and Y will produce the observed paths?

		X	Y	
	A	positive	positive	
	B	positive	negative	
	C	negative	positive	

[Turn over

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	D	negative	negative	
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Ans: A

downward force in downward field (force and field same direction) hence positive

21	A point charge $+Q$ is placed at the top left hand corner of a square as shown
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